

3. Identify the first and third quartiles.

Solution:

The first quartile corresponds to the lowest 25 percent of the ranked data, or -0.293 percent (and lower).

The third quartile corresponds to the lowest 75 percent of the ranked data, or 0.460 percent (and lower).

4. Identify the median.

Solution:

The median is the return for which 50 percent of the data lies on either side, which is 0.044 percent, the highest daily return in the 10th bin out of 20.

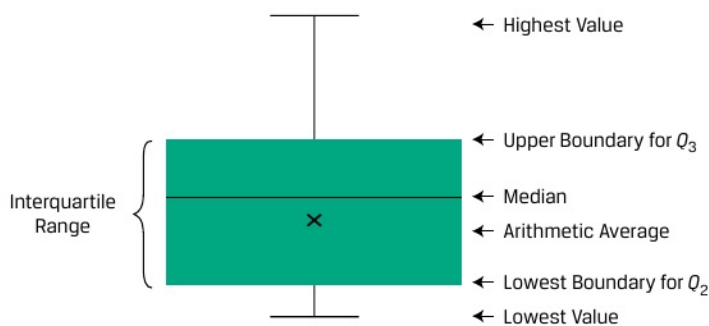
5. Calculate the interquartile range.

Solution:

The interquartile range is the difference between the third and first quartiles, 0.460 percent and -0.293 percent, or 0.753 percent.

One way to visualize the dispersion of data across quartiles is to use a diagram, such as a box and whisker chart. A **box and whisker plot** is shown in Exhibit 5. The “box” represents the lower bound of the second quartile and the upper bound of the third quartile, with the median or arithmetic average noted as a measure of central tendency of the entire distribution. The whiskers are the lines that run from the box and are bounded by the “fences,” which represent the lowest and highest values of the distribution.

Exhibit 5: Box and Whisker Plot



There are several variations for box and whisker displays. For example, for ease in detecting potential outliers, the fences of the whiskers may be a function of the interquartile range instead of the highest and lowest values like that in Exhibit 5.

In Exhibit 5, visually, the interquartile range is the height of the box and the fences are set at extremes. But another form of box and whisker plot typically uses 1.5 times the interquartile range for the fences. Thus, the upper fence is 1.5 times the interquartile range added to the upper bound of Q_3 , and the lower fence is 1.5 times the interquartile range subtracted from the lower bound of Q_2 . Observations beyond the fences (i.e., outliers) may also be displayed.

We can see the role of outliers in such a box and whisker plot using the EAA Equity Index daily returns, as shown in Exhibit 6.